THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 20	Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
2	**(c)** -14	Put $p(z) = 0$.
3	**(d)** $\frac{39}{65}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
4	**(d)** $49x^2 + 56xy + 16y^2$	Use the square identity.
5	**(a)** 6	Chord $= 2\sqrt{r^2 - d^2}$.
6	**(c)** 22	Arc $= \frac{\theta}{360} \times 2\pi r$.
7	**(b)** $\frac{1}{6}$	There are 36 equally likely outcomes.
8	**(b)** 4	$d = \frac{t_n - t_1}{n-1} = \frac{40}{10}$.
9	**(c)** 9 is divisible by 3 but not by 6	Counterexample for the converse.
10	**(c)** 87	$\frac{12 \times 89 + 63}{13} = 87$.
11	**(c)** 9	Pairs: $(360, 156) \to (204, 156) \to (156, 48) \to (108, 48) \to (60, 48) \to (48, 12) \to (36, 12) \to (24, 12) \to (12, 12) \to (12, 0)$.
12	**(c)** 150°	One part $= 360^\circ \div 12 = 30^\circ$.
13	**(d)** -1	Rewrite as $y = -x + \ldots$; slope $= -\frac{a}{b}$.
14	**(c)** 36	Surface area $\propto r^2$, so $6^2 = 36$.
15	**(a)** $(4.5,\ 4.5)$	Average the coordinates.
16	**(a)** 3	Slope $= \frac{y_2 - y_1}{x_2 - x_1}$.
17	**(d)** 4	$400 = 2^4 \times 5^2$
18	**(d)** $6x^2 + 19x + 15$	Multiply term by term.
19	**(a)** 5	$r^2 = h^2 + d^2$.
20	**(c)** 12	$h = \frac{2A}{b}$.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$d^2 = (4 - (1))^2 + (5 - (1))^2 = 25$ (1); $d = 5$ units (1).	1+1
22	$m = \frac{13-(4)}{5-(2)}$ (1); $m = 3$ (1).	1+1
23	Multiply by 10 and by 100 and subtract: $90x = 75$ (1); $x = \frac{5}{6}$ (1).	1+1
24	Multiply term by term (1); $= 6x^2 + 17x + 5$ (1).	1+1
25	Half-chord $= \sqrt{20^2 - 16^2} = 12$ cm (1); Chord $= 24$ cm (1).	1+1
26	$s = 32$, $A = 168$ cm^2 (1); $h = \frac{2 \times 168}{14} = 24$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$AB = 1 \times 5 = 5$ (direction $(3, 4)$); $BC = 2 \times 5 = 10$ (1); $AC = 15 = AB + BC$ (1); So the points are collinear (1).	1+1+1
28	$18 = m(2) + 12$ (1); $m = 3$ (1); x-intercept: $0 = 3x + 12$, $x = -4$ (1).	1+1+1
29	$1000x - x = 27$ (1); $999x = 27$ (1); $x = \frac{1}{37}$ (1).	1+1+1
30	$x^2 + \frac{1}{x^2} = (6)^2 - 2 = 34$ (1); $\left(x - \frac{1}{x}\right)^2 = 34 - 2 = 32$ (2).	1+2
31	Distances from the centre: 15 cm and 8 cm (1); Same side: subtract (1); Distance $= 7$ cm (1).	1+1+1
	OR: $s = 20$ cm (1); $s - a = 16$, $s - b = 12$, $s - c = 9$, $s - d = 3$ (1); $A = 72$ cm^2 (1).	1+1+1
32	$P(\text{a number less than } 3) = \frac{1}{3}$ (1); $P(\text{a number greater than } 4) = \frac{1}{3}$ (1); Favourable for either: 4, so $P = \frac{2}{3}$ (1).	1+1+1
	OR: $n = \frac{l-a}{d} + 1 = 25$ (1); $S = \frac{n}{2}(a + l)$ (1); $S = 2000$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$AB^2 = 9 + 16 = 25$ (1); $BC^2 = DA^2 = CD^2 = 25$: all sides equal 5 (1); $AC = 2 \times 3 = 6$ (1); $BD = 2 \times 4 = 8$ (1); $6 \neq 8$, so not a square (1).	1+1+1+1+1
34	Slope $= \frac{16-(8)}{3-(1)} = 4$ (1); $p(x) = 4x + 4$ (1); Zero: $x = -1$ (1); $p(5) = 24$ (2).	1+1+1+2
35	Let $x = 1.1\overline{6}$; $10x$ and $100x$ give $90x = 105$ (1); $x = \frac{7}{6}$ (1); Long division: $1.1\overline{6}$ (1); Matches the given decimal (2).	1+1+1+2
36	$x^2 - y^2 = (x + y)(x - y)$ (1); $= 12 \times 4 = 48$ (1); $4xy = (x + y)^2 - (x - y)^2 = 144 - 16$, so $xy = 32$ (1); $2(x^2 + y^2) = (x + y)^2 + (x - y)^2 = 160$ (1); $x^2 + y^2 = 80$ (1).	1+1+1+1+1
	OR: Half-chord $= 12$ cm (1); Distance $d = \sqrt{13^2 - 12^2} = 5$ cm (1); Area $= \frac{1}{2} \times 24 \times 5$ (1); $= 60$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
37	Plot area $= 1200$ m^2 (1); Pond area $= \frac{22}{7} \times 7^2 = 154$ m^2 (1); Remaining $= 1046$ m^2 (1); Cost $= 1046 \times 6$ (1); $=$ ₹6276 (1).	1+1+1+1+1
	OR: Sum 5: 4 outcomes; sum 9: 4 outcomes (1); $P(5) = \frac{1}{9}$ (1); $P(9) = \frac{1}{9}$ (1); $P(\text{either}) = \frac{8}{36} = \frac{2}{9}$ (2).	1+1+1+2
38	Lengths form an AP with $d = -4$ (1); Top rung $= 120 - 10 \times 4 = 80$ cm (1); $S = \frac{n}{2}(a + l)$ (1); $S = \frac{11}{2}(120 + 80)$ (1); Total 1100 cm (1).	1+1+1+1+1
	OR: Numbers $2n, 2n + 2, \ldots, 2n + 4$ (1); Sum $= 6n + 6 = 6(n + 1)$ (1); For $n = 4$: 30 (1); $30 = 6 \times 5$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERPIECE

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{6}$	Count favourable outcomes out of 36.
2	**(c)** 9	$a = t_n - (n-1)d = 69 - 60$.
3	**(d)** 6	6 is even.
4	**(a)** 85	$\frac{27 \times 84 - 58}{26} = 85$.
5	**(b)** 9	Pairs: $(420, 364) \to (364, 56) \to (308, 56) \to (252, 56) \to (196, 56) \to (140, 56) \to (84, 56) \to (56, 28) \to (28, 28) \to (28, 0)$.
6	**(c)** 55°	Let angles be x and $x + d$; they add to 180°.
7	**(a)** 14	Equal slopes between consecutive points.
8	**(a)** 125	Volume $\propto a^3$, so $5^3 = 125$.
9	**(c)** $(-1, -3)$	The centre is the midpoint of a diameter.
10	**(b)** -3	Put $y = 0$.
11	**(c)** $\frac{4}{33}$	$x = \frac{ab}{99}$, then reduce.
12	**(b)** 45	$a^2 + b^2 = (a-b)^2 + 2ab$.
13	**(a)** 58°	Isosceles triangle OAB.
14	**(c)** 12	$h = \frac{A}{b}$.
15	**(b)** $\frac{2}{3}$	Count the outcomes in either event, without double counting.
16	**(b)** 59	$5d = 20$ so $d = 4$; then $t_{15} = t_9 + 6d$.
17	**(d)** 4	$n = 4$ gives 121, which is composite.
18	**(d)** 64	Sum of the extra numbers $= 70 \times 60 - 40 \times 73 = 1280$; divide by 20.
19	**(c)** 23	Multiples of 3: 16; of 5: 10; of both: 3.
20	**(a)** 480 cm^2	Other side $= 30$ cm; area $= 16 \times 30$.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Outcomes with sum 6: (1,5), (2,4), (3,3), (4,2), (5,1) (1); $P = \frac{5}{36} = \frac{5}{36}$ (1).	1+1
22	$(q-p)d = 30$ (1); $d = 6$ (1).	1+1
23	Converse: If a number has an odd number of factors, then it is a perfect square. (1); it is true (1).	1+1
24	Sum $= 32 \times 44 = 1408$; Remaining sum $= 1408 - 106 = 1302$ (1); New mean $= \frac{1302}{31} = 42$ (1).	1+1
25	$(106, 44) \to (62, 44) \to (44, 18) \to (26, 18) \to (18, 8) \to (10, 8) \to (8, 2) \to (6, 2) \to (4, 2) \to (2, 2) \to (2, 0)$; gcd $= 2$ (1); LCM $= \frac{106 \times 44}{2} = 2332$ (1).	1+1
26	Let the angles be x and $x + 20$; adjacent angles sum to 180°; $2x + 20 = 180$ gives $x = 80$ (1); Angles: $80^\circ, 100^\circ, 80^\circ, 100^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Converse: If the square of the longest side of a triangle equals the sum of the squares of the other two sides, then the triangle is right-angled. (1); The converse is also true (1); Both the proposition and its converse hold, so we may write the two as a single 'if and only if' statement (1).	1+1+1
28	Let the number of remaining students be n; $21 \times 56 + 36n = 41(21 + n)$ (1); $n(36 - 41) = 21(41 - 56)$ (1); $n = 63$ (1).	1+1+1
29	divisors(84) $= [1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84]$, divisors(68) $= [1, 2, 4, 17, 34, 68]$; Greatest common divisor from the lists $= 4$ (1); Subtraction: $(84, 68) \to (68, 16) \to (52, 16) \to (36, 16) \to (20, 16) \to (16, 4) \to (12, 4) \to (8, 4) \to (4, 4) \to (4, 0)$, 9 subtractions (1); gcd $= 4$ by both methods (1).	1+1+1

30	Area $= \frac{1}{2}d^2 = \frac{1}{2} \times 18^2 = 162$ cm^2 (1); Side$^2 = 162$ (1); Side $= 9\sqrt{2}$ cm (1).	1+1+1
31	x-intercept: put $y = 0$: $8x = 32$, so $x = 4$ (1); y-intercept: put $x = 0$: $4y = 32$, so $y = 8$ (1); Area $= \frac{1}{2} \times 4 \times 8 = 16$ square units (1).	1+1+1
	OR: $l = \sqrt{r^2 + h^2} = 17.5$ cm (1); CSA $= \pi rl = 577.5$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 1617$ cm^3 (1).	1+1+1
32	$(2 - k)^2 + 16 = 25$ (1); $(2 - k)^2 = 9$ (1); $k = -1$ or $k = 5$ (1).	1+1+1
	OR: Points $(0, 5)$, $(1, 1)$, $(2, -3)$ (1); Slope $= \frac{-3-(5)}{2-0}$ (1); $= -4$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Totals: $1886, 2200, 1920$ (1); Grand total $= 6006$ for 78 students (2); Average $= 77$ (2).	1+2+2
34	Values of r: 2, 23, 234, 2341 (1); Final $r = 2341$ (2); It reverses the digits (2).	1+2+2
35	In $\triangle ABC$, $AB^2 + BC^2 = 121 + 36 = 157$ (as $BC = AD = 6$) (1); This equals AC^2, so $\angle B = 90^\circ$ (converse of the Baudhāyana–Pythagoras theorem) (1); A parallelogram with one right angle has all angles 90° (1); Hence ABCD is a rectangle (2).	1+1+1+2
36	Slope $= \dfrac{40}{20} = 2$ (1); Intercept $= 50 - 2 \times 20 = 10$, so $y = 2x + 10$ (1); $x = 128$: $y = 266$ (1); $286 = 2x + 10$ gives $x = 138$ (2).	1+1+1+2
	OR: Volume of cylinder $= \pi r^2 \times 24$ (1); Volume of one cone $= \frac{1}{3}\pi r^2 \times 1$ (2); $n = \dfrac{\pi r^2 \times 24}{\frac{1}{3}\pi r^2 \times 1} = \dfrac{3 \times 24}{1} = 72$ (2).	1+2+2
37	$D = A + C - B = (2, 5)$ (1); Midpoint of $AC = (3.5, 3)$ (1); Midpoint of $BD = (3.5, 3)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1).	1+1+1+1+1
	OR: $E(n) = 150n + 5000$ (1); $E(20) = 8000$ (1); $150n + 5000 = 14000$, $n = 60$ (1); Extra items $= \frac{1200}{150} = 8$ (2).	1+1+1+2
38	$(3 + \sqrt{5})^2 = 14 + 6\sqrt{5}$ (1); $(3 - \sqrt{5})^2 = 14 - 6\sqrt{5}$ (1); Sum $= 28$ (1); The surd terms cancel (1); Result is rational (1).	1+1+1+1+1
	OR: $(a + b)^2 = a^2 + b^2 + 2ab$ (1); $81 = a^2 + b^2 + 28$ (1); $a^2 + b^2 = 53$ (1); $(a - b)^2 = (a + b)^2 - 4ab = 81 - 56 = 25$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	Equal slopes: $-\frac{4}{2} = -\frac{k}{4}$, so $k = 8$.
2	**(b)** $1 : 216$	Ratio of volumes $= r_1^3 : r_2^3 = 1 : 216$.
3	**(b)** $(3, 0)$	Set $PA = PB$ with $P = (x, 0)$.
4	**(d)** 4	Substitute y and solve for x.
5	**(c)** 6	Find how many digits repeat.
6	**(a)** 7	Square both sides.
7	**(d)** 72°	$\angle A + \angle C = 180^\circ$.
8	**(c)** 52	Side $= \sqrt{A}$.
9	**(a)** $\frac{7}{12}$	Both favourable and total counts increase.
10	**(a)** 182	$2 + 4 + \cdots + 2n = n(n + 1)$.
11	**(b)** 12	List the factors of 72 in pairs; the count is 12 (even, so 72 is not a perfect square).
12	**(a)** 24	$12(73 - 71) = n_2(74 - 73)$.
13	**(d)** 18	The algorithm adds the digits.
14	**(c)** 52 cm	Side $= 13$ cm, perimeter $= 4 \times 13$.
15	**(d)** 3	Intercepts 2 and 3; area $= \frac{1}{2} \times 2 \times 3$.
16	**(c)** 924 cm^2	$\pi r(l + r)$ with the given values.
17	**(a)** Quadrant III	Use the sign conditions on the coordinates.
18	**(a)** $8x - 5$	Add like terms.
19	**(d)** 6	Multiply under the root.
20	**(a)** 1006009	Write $1003 = 1000 + 3$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Rate $= \dfrac{163 - 82}{12 - 3} = 9$ per km; Fixed charge $= 82 - 9 \times 3 = 55$ (1); Fare for 23 km $= 9 \times 23 + 55$; $=$ ₹262 (1).	1+1
22	$\pi r^2 h = 2079$ gives $r^2 = \dfrac{2079 \times 7}{22 \times 6} = 110.25$; $r = 10.5$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 10.5 \times 6 = 396$ cm^2 (1).	1+1
23	$\left(\frac{3+(6)}{2}, \frac{-4+(7)}{2}\right)$ (1); Midpoint $= (4.5,\ 1.5)$ (1).	1+1
24	$p - q = 3x + 9$ (1); Zero $x = -3$ (1).	1+1
25	Long division of 5 by 22 (1); $= 0.2\overline{27}$, block 27 (1).	1+1
26	$x^2 + y^2 = (x + y)^2 - 2xy$ (1); $= 121 - 56 = 65$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$x = 1 + 0.2\overline{3}$ (1); $0.2\overline{3} = \frac{7}{30}$ (1); $x = \frac{37}{30}$ (1).	1+1+1
28	$a^2 + b^2 = 4^2 + 2(21)$ (1); $= 58$ (1); $(a + b)^2 = 16 + 4(21) = 100$ (1).	1+1+1
29	$\angle A + \angle C = 180^\circ$; $5x = 170$ (1); $x = 34$ (1); $\angle A = 122^\circ$, $\angle C = 58^\circ$ (1).	1+1+1
30	Perimeter $= 2 \times 30 + 14 + \frac{1}{2}\pi \times 14 = 74 + 22$; $= 96$ cm (1); Area $= 420 + \frac{1}{2}\pi \left(\frac{14}{2}\right)^2$ (1); $= 497$ cm^2 (1).	1+1+1
31	Likes tea $= 70 + 30 = 100$; $P(\text{tea}) = \frac{100}{200} = \frac{1}{2}$ (1); Neither $= 200 - 70 - 50 - 30 = 50$ (1); $P(\text{neither}) = \frac{1}{4}$ (1).	1+1+1
	OR: $d = 5$ (1); $a = 6$ (1); $t_{20} = 101$ (1).	1+1+1
32	A false statement needs only one counterexample (1); Take 18 (1); 18 is divisible by 3 and 9, not 27, so the statement is false (1).	1+1+1

	OR: Total score of all $= 68 \times 99 = 6732$; Total score of the first group $= 45 \times 80 = 3600$ (1); Total score of the additional group $= 3132$ (1); Average $= \dfrac{3132}{54} = 58$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$d_1 = \sqrt{13^2 - 5^2} = 12$ cm (1); $d_2 = \sqrt{13^2 - 12^2} = 5$ cm (1); The chord of length 24 cm is nearer to the centre (1); A longer chord lies nearer to the centre (2).	1+1+1+2
34	$s = 18$ m (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 60$ m^2 (1); Cost $= 60 \times 5$ (1); $=$ ₹300 (1); Units stated (1).	1+1+1+1+1
35	$P(\text{defective}) = \frac{8}{200} = \frac{1}{25}$ (1); Expected defective $= 5000 \times \frac{1}{25}$ (1); $= 200$ (1); Units stated (2).	1+1+1+2
36	Each year multiplies by $110/100$ (1); Hence a GP with this common ratio (1); Amount after n years $= P(1 + \frac{r}{100})^n$ (1); $= 10000 \times (1.1)^3$ (1); Amount ₹13310 (1).	1+1+1+1+1
	OR: $47 - (4 + 7) = 36$ (1); $36 = 4 \times 9$ (1); General number $10a + b$ (1); $10a + b - (a + b) = 9a$ (1); A multiple of 9 (1).	1+1+1+1+1
37	Pure part in the first $= 510 \times \frac{22}{100} = 112.2$ mL (1); Pure part in the second $= 102$ mL (1); Total $= 214.2$ mL in 1190 mL (1); Concentration $= 18\%$ (2).	1+1+1+2
	OR: Sequence: 9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 19$ (3).	2+3
38	Midpoint theorem on each pair of sides (1); $DE = \frac{1}{2}AC$, $EF = \frac{1}{2}AB$, $FD = \frac{1}{2}BC$ (2); Perimeter of $\triangle DEF = \frac{1}{2}(82) = 41$ cm (2).	1+2+2
	OR: $y = mx + c$ with $m = 5$: $25 = 5(4) + c$ gives $c = 5$ (1); Equation: $y = 5x + 5$ (1); x-intercept: $0 = 5x + 5$ gives $x = -1$ (1); At $x = 2$: $y = 15$ (2).	1+1+1+2

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